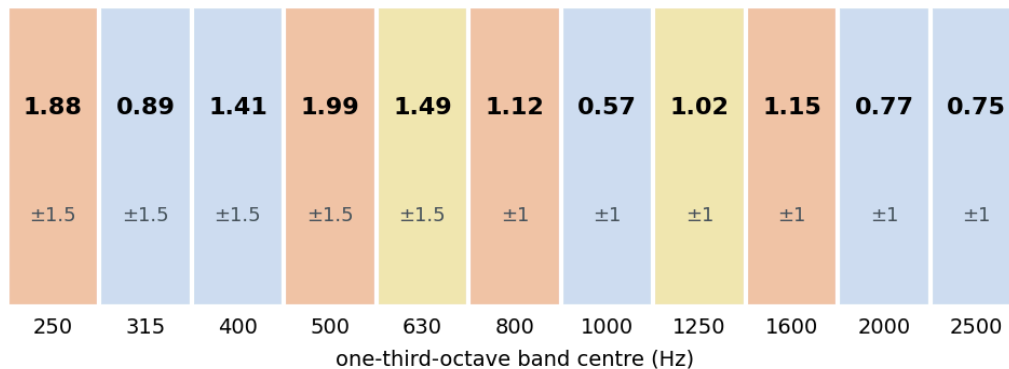


Chamber report — final configuration

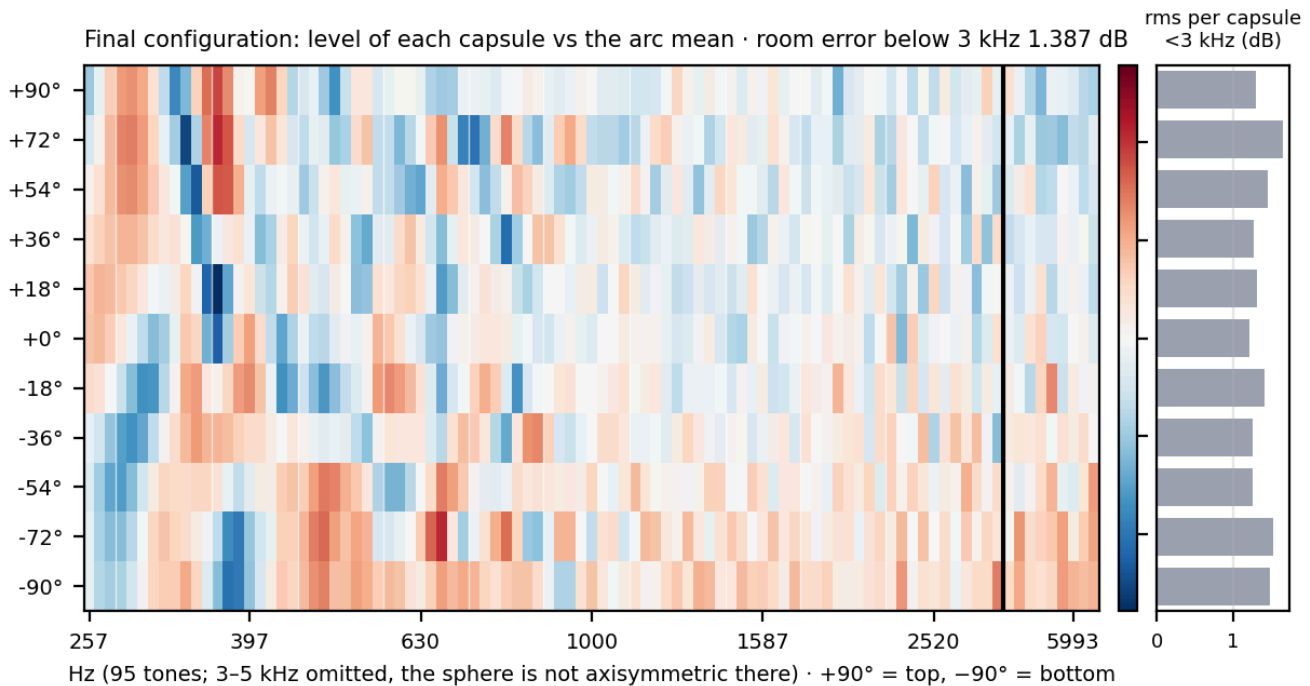
SoundVisualizer · 2026-10-01 · 11 calibrated UMIK-2 capsules on a 0.84 m arc · loudspeaker sphere on the axis · 95 tones, 257 Hz–6.35 kHz

Final configuration (carpet-reordered, accepted by Adam): room error **1.387 dB** against **2.054 dB** for the worst bare room (-32 %) and 1.380 for the untouched start. By the ISO 3745 test applied band by band it is **qualified from the 2000 Hz band up**; below that, 4 of 11 bands are clearly over the limit and 2 are marginal. **Pure tones are qualified in no band**: 65 % of tone × capsule cells are inside the limit.

Worst capsule vs arc mean per band (dB), against the ISO 3745 limit · orange = over, yellow = within ± 0.09 of it



Final configuration: level of each capsule vs the arc mean · room error below 3 kHz 1.387 dB

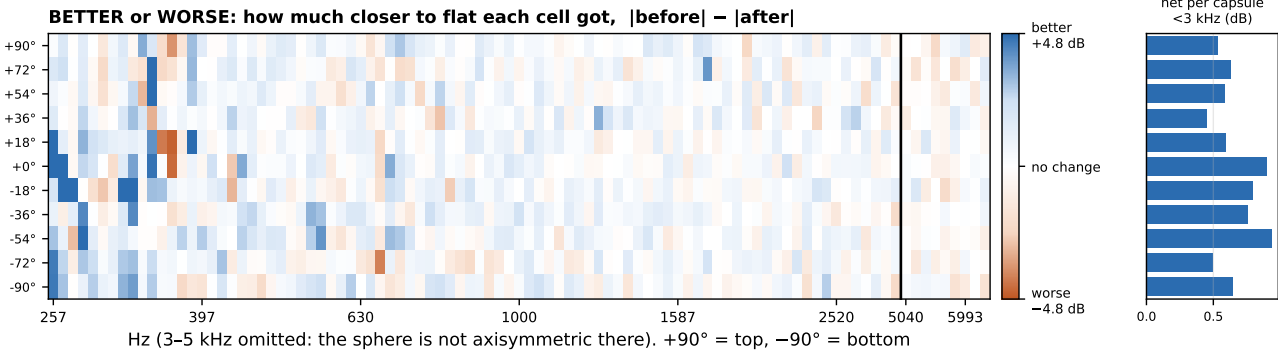
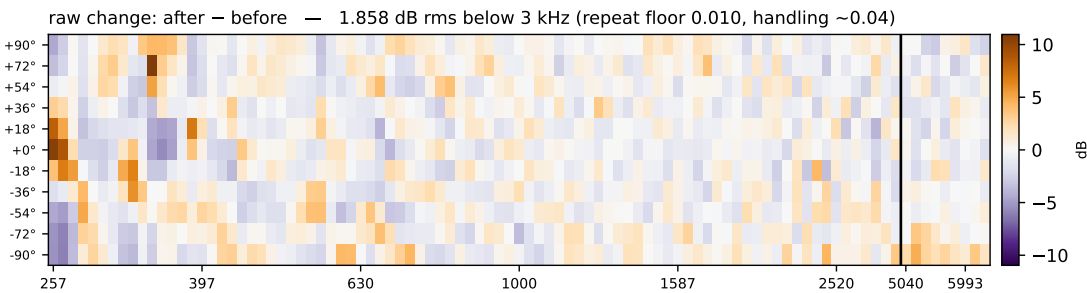
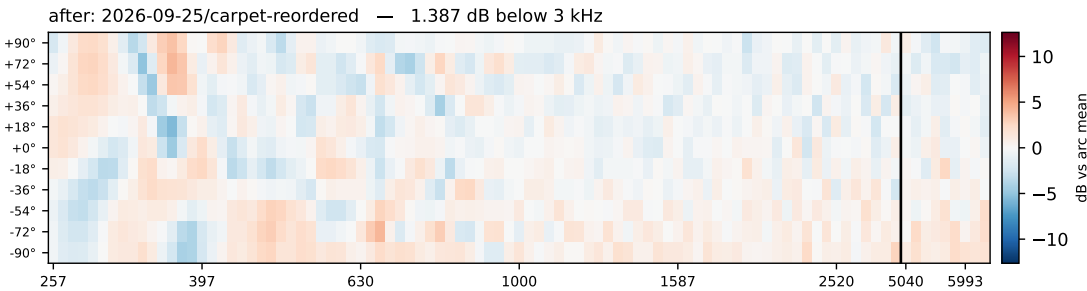
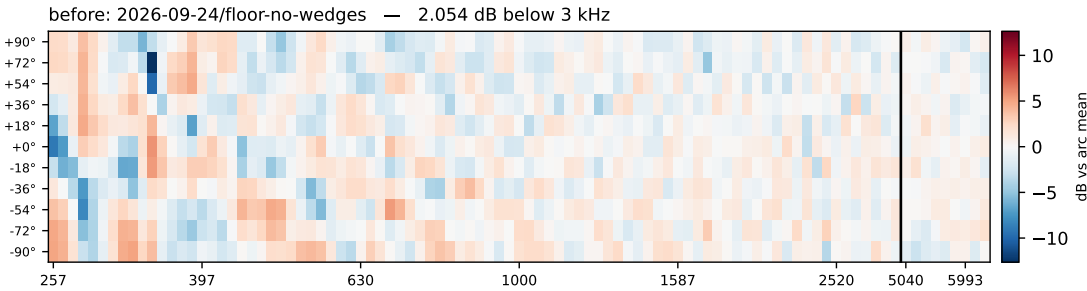
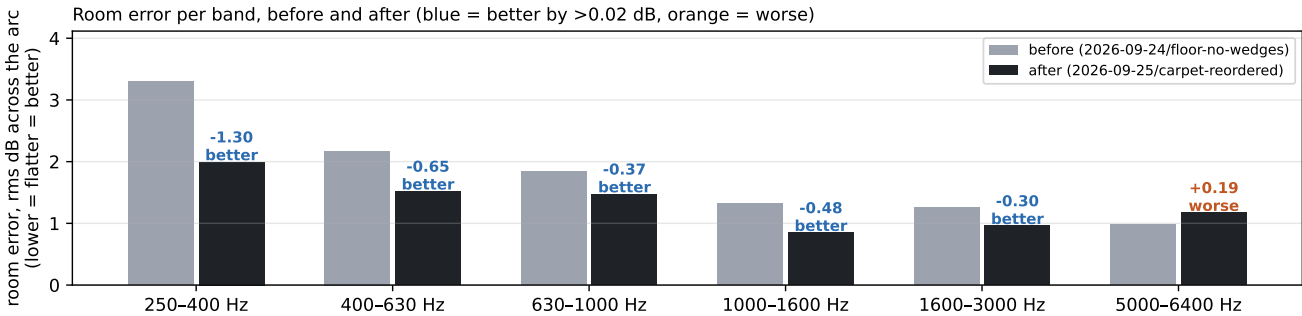


How to read: each cell of the map is one capsule at one tone; blue = quieter than the arc mean, red = louder. In an ideal free field every capsule would read the same, so the map would be white. The strip above gives, per band, the worst capsule's band-averaged deviation; the limits are the ISO 3745 anechoic ones (± 1.5 dB to 630 Hz, ± 1.0 dB from 800 Hz).

- Worst bands by room error: 250–400 Hz (1.99 dB) and 400–630 Hz (1.52); best: 1–1.6 kHz (0.85).
- Reproducibility: the same configuration, re-measured on 2026-09-30, scores 1.399 (map differs by 0.13 dB rms).

Final configuration vs the empty floor (wedges off): room error below 3 kHz 2.054 → 1.387 dB (-0.668, better)

2026-09-24/floor-no-wedges → 2026-09-25/carpet-reordered · below 3 kHz, 600 of 946 cells moved >0.05 dB toward flat and 308 away from it · 95 tones · 11 calibrated capsules



Evaluation by the method of the chamber papers

The method

Chambers are qualified one one-third-octave band at a time. Each band's measured level is compared with the free-field ideal and must stay inside a tolerance; the chamber is then reported as qualified between two bands, and its cut-off is the lowest band above which everything passes. Noise and pure tones are reported separately.

"TABLE I. Maximum allowable difference in anechoic rooms between measured and theoretical free-field levels per ISO 3745 and ANSI S12.35."
Cunefare et al. 2003, J. Acoust. Soc. Am. 113(2), p. 882 – ±1.5 dB up to 630 Hz, ±1.0 dB from 800 to 5000 Hz

"semi-anechoic chamber is qualified with ISO 3745 on the 1/3 frequency band between 160 Hz- 4000 Hz. Cut-off frequency of the chamber is 160 Hz."
Kayhan 2008, METU thesis, p. 63

"The chamber could satisfy the ISO tolerances when using random noise but failed to qualify when using pure tones."
Nash 2019, Proc. 23rd ICA, p. 1343

Our adaptation. The standard moves a microphone away from the source and compares the decay with 1/r. We have eleven fixed positions around an axisymmetric source, so the reference is the arc mean. **Band level:** each capsule's mean over the 8–9 tones in the band (noise-like use); the worst capsule must be inside the limit. **Pure tones:** the share of single tone × capsule cells inside the limit. A band within ±0.09 dB of its limit is marginal: that is how far the statistic moved between the final configuration and its rerun five days later.

Result, band by band (worst capsule, dB)

BAND	LIMIT	TONES	WORST BARE ROOM	TONES IN LIMIT	FINAL CONFIGURATION	TONES IN LIMIT
250 Hz	±1.5	3	5.71	36 %	1.88	52 %
315 Hz	±1.5	8	2.60	28 %	0.89	56 %
400 Hz	±1.5	8	2.05	36 %	1.41	59 %
500 Hz	±1.5	8	2.25	44 %	1.99	59 %
630 Hz	±1.5	8	1.64	52 %	1.49	61 %
800 Hz	±1	8	1.42	42 %	1.12	55 %
1000 Hz	±1	9	0.89	41 %	0.57	73 %
1250 Hz	±1	8	0.72	61 %	1.02	73 %
1600 Hz	±1	8	1.39	47 %	1.15	76 %
2000 Hz	±1	8	0.98	65 %	0.77	82 %
2500 Hz	±1	8	0.81	53 %	0.75	65 %

The cut-off is set by the highest failing band, so one band can move it; read it together with the lists. Shading: blue = inside the limit, yellow = marginal, orange = over. Worst bare room = floor-no-wedges: the wedges off, the flat foam base left – the nearest measured state to an empty room.

Scorecard

MEASURE	WORST BARE ROOM	UNTOUCHED START	FINAL CONFIGURATION
Room error below 3 kHz (dB)	2.054	1.380	1.387
Band-level qualified from	2000 Hz	none	2000 Hz
Bands clearly over the limit (of 11)	7: 250, 315, 400, 500, 630, 800, 1600 Hz	4: 250, 315, 500, 2500 Hz	4: 250, 500, 800, 1600 Hz
Bands marginal	1: 2000 Hz	2: 400, 1600 Hz	2: 630, 1250 Hz
Pure-tone cells inside the limit	47 %	66 %	65 %

Where that puts the chamber

- Against the worst case:** room error 32 % lower, bands clearly over the limit 7 → 4, pure-tone cells inside the limit 47 → 65 %. The cut-off is the same, 2000 Hz. Not better everywhere: at 1250 Hz the bare room is the better one (0.72 vs 1.02).
- Against the untouched start:** no gain in the headline (1.380 → 1.387) and the same number of failing bands (4), in different places: the work fixed 315, 2500 Hz and broke 800, 1600 Hz.
- Against other chambers:** published small rotor chambers report cut-offs of 63–275 Hz (29 facilities, chamber-fighting-guide.pdf §01); METU, in the quote above, 160 Hz. Ours is the 2000 Hz band, 7–32 times higher, by a test that is easier in one respect (a reference that is the arc mean, not the ideal) and harder in another (single worst capsule of eleven).

What the method cannot say

- Deviation from the arc mean is blind to an error shared by all eleven positions (a uniform reflection). The ISO traverse has not been run here, so this is a ranking of bands, not a qualification certificate.
- The grid starts at 257 Hz and the blade tone sits at 215–258 Hz; the 250 Hz band has only 3 tones. 3–5 kHz is excluded (the sphere's rocking mode); 5–6.4 kHz is indicative.
- The final configuration's source sits in a different position from the start's (the tilt flipped on 25 Sep); with the tilt fitted out its score is lower than the start's. The scorecard compares whole setups, not individual treatments.